

PROFESSIONAL SUMMARY

Software Engineer focused on backend and full-stack development, with a track record of delivering high-impact solutions for complex, high-stakes systems, driving rapid delivery, product reliability, and customer trust.

EXPERIENCE

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| <b>Software Development Engineer I</b><br><i>SAIF Corporation</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | June 2025 – Dec 2025<br><i>Salem, OR</i>     |
| <ul style="list-style-type: none"><li>Engineered corrections to the short-rate cancellations in billing engine, implementing transactional logic using Gosu to prevent erroneous refunds and safeguard \$100k+ in annual revenue.</li><li>Refactored a legacy VB.NET actuarial regulatory reporting tool into C#, introducing parallel bulk processing and unit test coverage to reduce turnaround time from days to hours.</li><li>Unblocked new customer acquisition by resolving an address validation issue in the issuance workflow that prevented foreign customers from purchasing policies, enabling seamless onboarding and avoiding manual workarounds.</li><li>Developed BDD scenarios with Cucumber/Gherkin to validate core business workflows, eliminating one week of manual validation and ensuring on-time regulatory compliance for high-stakes releases.</li></ul>                                            |                                              |
| <b>Reliability Engineer Intern</b><br><i>MKS Instruments, Inc.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Apr 2024 – Sept 2024<br><i>Beaverton, OR</i> |
| <ul style="list-style-type: none"><li>Extended C# .NET test infrastructure for a precision motion control system, leveraging dependency injection to validate against both simulated and physical hardware and improving test result reporting.</li><li>Designed a field failure tracking system and data consolidation pipeline for laser components, processing 10k+ records spanning decades to deliver reliability insights and RCA to leadership, informing warranty projections.</li><li>Engineered Python CLI tool to compress and archive AI computer vision defect detection data from the manufacturing floor, reducing storage needs by 90% and saving \$3000+ annually in hardware costs.</li><li>Automated quality metrics reporting across cross-functional teams using Azure dataflows and Power BI backed by Azure Data Lake, eliminating manual entry errors and saving leadership 10+ hours monthly.</li></ul> |                                              |

SKILLS

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| <b>Languages:</b> Python, C#, JavaScript/TypeScript, Gosu (JVM), SQL, HTML/CSS, Bash                |
| <b>Web &amp; Backend:</b> React, Express.js, Django, Node.js, .NET, REST APIs, Docker, Git          |
| <b>Data &amp; Cloud:</b> MySQL, PostgreSQL, NoSQL (MongoDB, LMDB), Redis, AWS (EC2, S3, IAM)        |
| <b>Practices:</b> Agile/Scrum, CI/CD (Azure DevOps), TDD/BDD, Infrastructure-as-Code, System Design |
| <b>Generative AI:</b> Claude Code, Cursor, AI-Augmented Development, Agentic Development            |

EDUCATION

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| <b>Oregon State University</b><br><i>B.S. in Computer Science, AI Focus; Minor in Mathematics</i>                                                                                                            | Corvallis, OR<br><i>Sept 2021 – June 2025</i> |
| <ul style="list-style-type: none"><li><b>GPA:</b> 3.94; Graduated <i>Summa Cum Laude</i></li><li><b>Activities:</b> AI Club President, Triathlon Club Vice President, Treasurer, and Race Director</li></ul> |                                               |

PROJECTS

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| <b>OpenResponse</b><br>Real-time student engagement and analytics platform                                                                                                                                                                                                                                                                                                                                                                                                                                            | Sept 2024 – June 2025<br><i>React, JavaScript, MySQL, Redis, Docker, AWS</i> |
| <ul style="list-style-type: none"><li>Led a 5-developer Agile team building a scalable React/Express application on AWS, directing sprints and technical architecture while implementing accessibility via UX prototypes and automated testing for WCAG compliance.</li><li>Architected a distributed backend system and deployed it to scalable, containerized AWS infrastructure supporting up to 10k+ concurrent users under load testing, and built CI/CD pipelines, strengthening release reliability.</li></ul> |                                                                              |
| <b>AI-Powered Hybrid Search Engine</b><br>High-performance file search engine with hybrid document retrieval                                                                                                                                                                                                                                                                                                                                                                                                          | Feb 2025 – Mar 2025<br><i>Python, Redis, SQLite, FAISS, REST API</i>         |
| <ul style="list-style-type: none"><li>Engineered a hybrid search engine combining keyword and vector-based semantic retrieval using custom-trained embeddings and FAISS, achieving 0.06s latency and 10x faster results than Windows Search for 100k+ files.</li><li>Designed a modular, event-driven AI indexing pipeline with a Redis worker queue, SQLite metadata store, and enabling incremental updates and near-real-time synchronization of file changes via semantic embeddings.</li></ul>                   |                                                                              |